

Drug Discovery AI Agent: A Retrieval-Based AI System with Persistent Memory, Dynamic Orchestration, and PubChem Skill Integration

Course: Deep Learning and Life Science

Project Title: Drug Discovery AI Agent

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GitHub Repository: https://github.com/BoolyLutoonom/ai-agent-assignment/tree/main/DL_Assignment

Abstract

This short paper explains the design, implementation, and evaluation of our project titled **Drug Discovery AI Agent**. The project was developed as a domain-specific AI agent for answering drug-discovery-related questions. Drug discovery is the scientific process of identifying and developing new medicines, and it includes stages such as target identification, compound screening, lead optimization, ADME analysis, toxicity testing, and clinical development. Our system was designed to demonstrate how an AI agent can support learning and information retrieval in this field. The project includes 25 structured drug-discovery knowledge datasets, 30 benchmark questions, a multi-level persistent memory system, advanced dynamic orchestration, and PubChem chemical data integration. The system runs by default in retrieval-based mode, which allows it to work without requiring a large local language model. It also supports optional local LLM use through the Phi-3-mini-4k-instruct model if the model is manually added. The project was tested successfully through validation, benchmark evaluation, orchestration demo, and memory demo scripts. The benchmark evaluation achieved an average score of 0.889, with all 30 benchmark questions passing the evaluation threshold.

Keywords: Drug discovery, AI agent, retrieval-based question answering, persistent memory, dynamic orchestration, PubChem, ADME, pharmacokinetics, pharmacodynamics

1. Introduction

Drug discovery is one of the most important areas in modern biomedical science because it focuses on finding and developing new medicines for diseases. The process usually involves identifying a biological target, screening compounds, optimizing promising molecules, testing drug-like properties, checking toxicity, and validating results through preclinical and clinical development. This process is scientifically complex, time-consuming, and expensive. Because of this, artificial intelligence is increasingly being used in pharmaceutical research to help organize knowledge, retrieve chemical information, support compound analysis, and improve early-stage decision-making.

The purpose of this project was to build a small but complete AI-agent prototype for the drug-discovery domain. Instead of creating a simple chatbot, we developed a system that can retrieve structured knowledge, remember previous user context, use an external chemical database skill, and dynamically select the correct workflow based on the user's question. This makes the system more advanced than a normal chatbot because it is not limited to static answers. It can decide whether to use the knowledge base, memory system, PubChem skill, or a combination of these components.

The project was designed for educational and demonstration purposes. It does not replace expert scientific or medical judgment. However, it shows how AI-agent concepts can be applied to a real life-sciences problem. The project also demonstrates important AI-agent features such as retrieval-based answering, persistent memory, tool use, orchestration, evaluation, and logging.

2. Project Objectives

The main objective of this project was to develop a working AI agent for drug-discovery question answering. The system was designed to answer both conceptual and tool-based questions. For example, it can explain concepts such as ADME, molecular docking, pharmacokinetics, pharmacodynamics, lead optimization, and virtual screening. It can also answer compound-related questions using PubChem, such as retrieving the molecular weight or formula of aspirin.

The specific objectives were:

1. To build a structured knowledge base for drug-discovery concepts.
2. To create a retrieval-based AI question-answering system.
3. To implement a multi-level persistent memory system.
4. To implement advanced dynamic orchestration for route selection.
5. To integrate a PubChem skill for chemical compound lookup.
6. To evaluate the system using benchmark questions.
7. To provide documentation, logs, test results, and a GitHub submission.

3. Selected Topic and Modules

The selected project topic was **drug-discovery**. This topic was chosen because it connects artificial intelligence with chemistry, biology, healthcare, and pharmaceutical research. It is also a suitable domain for an AI agent because drug-discovery questions often require both conceptual knowledge and chemical data lookup.

The selected optional modules were:

Module	Level	Purpose
Multi-level Persistent Memory System	Lv.1	Allows the system to remember and recall user context

Module	Level	Purpose
Advanced Dynamic Orchestration	Lv.3	Allows the system to choose the correct workflow dynamically
PubChem MCP/SKILL Integration	Lv.1	Allows the system to retrieve chemical compound information
Total Difficulty	5	Meets the assignment requirement

These modules were selected because they make the project more agentic. A basic chatbot usually only responds to user input, but our system can decide which component should handle the question. For example, a memory-based question is routed to memory, a compound lookup question is routed to PubChem, and a conceptual question is routed to the knowledge base.

4. System Architecture

The system architecture is based on a modular design. The main application is launched through `app.py`, while the core logic is organized inside the `src/` folder. The project also includes separate folders for knowledge content, benchmark questions, memory files, documentation, reports, and logs.

The general workflow is:

```

User Question
  ↓
Dynamic Orchestrator
  ↓
Route Selection
  ↓
Knowledge Retrieval / Memory Retrieval / PubChem Skill
  ↓
Answer Construction
  ↓
Logs and Reports

```

When the user asks a question, the orchestrator first analyzes the type of question. If the question is about a drug-discovery concept, the system retrieves relevant information from the structured knowledge base. If the question asks about a chemical compound, the system uses the PubChem skill. If the question asks about earlier conversation history, the system uses persistent memory. This workflow allows the system to behave more like an AI agent rather than a simple chatbot.

5. Knowledge Base Design

The project includes **25 structured drug-discovery knowledge datasets**. Each dataset follows a consistent structure with content, keywords, and source information. The knowledge base covers important topics in drug discovery, including:

- target identification,
- target validation,
- compound screening,
- lead optimization,
- molecular docking,
- virtual screening,
- ADME,
- toxicity testing,
- pharmacokinetics,
- pharmacodynamics,
- QSAR,
- drug repurposing,
- clinical trials,
- regulatory approval,
- AI explainability,
- PubChem compound lookup.

This structured knowledge design makes the system more reliable because it answers questions using prepared domain-specific material. It also makes the project easier to validate because each knowledge asset is stored in a clear format.

6. Persistent Memory System

The memory system was designed to show that the agent can remember information across multiple interactions. It includes three levels of memory:

1. **Working memory:** stores recent conversation turns.
2. **Episodic memory:** stores previous interactions in a persistent file.
3. **Semantic memory:** stores important entities and relationships.

For example, if a user says, “Remember that my project focuses on ADME and pharmacokinetics,” the system can store that information. Later, if the user asks, “What did I ask earlier about ADME?”, the system can recall the earlier context. This demonstrates that the agent is not only answering isolated questions but also maintaining user context.

The memory system was tested using `demo_memory.py`. The test confirmed that memory recall works even after another unrelated question is asked in between.

7. Advanced Dynamic Orchestration

Advanced dynamic orchestration is a key feature of the project. It allows the system to choose the correct workflow based on the question. This is important because not all questions should be answered in the same way.

For example:

User Question	System Route
What is ADME?	Knowledge retrieval
What is the molecular weight of aspirin?	PubChem skill
What did I ask earlier about ADME?	Memory retrieval
Why is a docking score not enough to approve a drug?	Knowledge retrieval and scientific explanation
What workflow is needed for a complex compound question?	Dynamic orchestration with possible multi-step routing

The orchestration module also creates trace logs. These logs show how the system selected the route and which component was used. This is useful for evaluation because it provides evidence that the system is making workflow decisions rather than simply returning fixed answers.

8. PubChem Skill Integration

The PubChem skill allows the agent to retrieve real chemical information about compounds. This is important in drug discovery because chemical properties should not be guessed. The skill can provide information such as molecular formula, molecular weight, InChIKey, and IUPAC name.

For example, when the user asks:

What is the molecular weight of aspirin?

The system can respond with information such as:

PubChem reports for aspirin: molecular formula C₉H₈O₄, molecular weight 180.16, InChIKey BSYNRYMUTXBXSQ-UHFFFAOYSA-N, and IUPAC name 2-acetyloxybenzoic acid.

This demonstrates external tool use, which is one of the main differences between a basic chatbot and an AI agent.

9. Evaluation and Results

The project was tested after cloning the GitHub repository into a fresh local folder. This was done to make sure that the uploaded submission works independently and does not depend on hidden local files.

The following commands were used:

```
python validate_submission.py
python evaluate_benchmark.py
python demo_orchestration_cases.py
python demo_memory.py
```

The validation result was:

```
{
  "knowledge_sets": 25,
  "benchmark_questions": 30,
  "errors": []
}
```

This shows that the project structure is correct and that the required knowledge and benchmark files are available.

The benchmark result was:

```
{
  "question_count": 30,
  "average_score": 0.889,
  "passed_at_0_34": 30
}
```

This shows that all 30 benchmark questions passed the evaluation threshold. The orchestration demo successfully generated `reports/orchestration_demo_traces.json`, and the memory demo confirmed that persistent memory recall works.

10. Default No-Model Mode and Optional LLM Support

The project runs by default without the Phi-3 model. This decision was made because the model file is large and should not be uploaded to GitHub. The default retrieval-based mode makes the system easier to run, test, and evaluate.

The optional model file is:

Phi-3-mini-4k-instruct.Q4_0.gguf

If needed, this model can be placed inside the `models/` folder to enable optional local LLM mode. However, the main assignment features work without the model because the system uses structured knowledge retrieval, memory, orchestration, and PubChem skill integration.

11. Division of Labor

Both team members contributed equally to the project. The work was divided in a balanced way so that both members participated in technical development, documentation, testing, and presentation preparation.

Team Member	Main Responsibilities	Contribution
Niteesh Kunwar	Project planning, AI-agent architecture, implementation of retrieval workflow, memory system testing, orchestration demo, GitHub submission, and final demo preparation	50%
Sekandari Mohammad Jamal Naser	Drug-discovery topic research, terminology explanation, knowledge asset review, benchmark question review, documentation support, presentation preparation, and final report review	50%

The team worked collaboratively to ensure that the final project included complete code, documentation, benchmark results, demo logs, and presentation material. Niteesh Kunwar focused more on the technical implementation and demonstration, while Sekandari Mohammad Jamal Naser focused more on domain explanation, documentation, and presentation support. Together, both members contributed to the final project submission.

12. Discussion

The project demonstrates the difference between a normal chatbot and an AI agent. A chatbot usually gives direct answers, but an AI agent can use memory, tools, and decision-making workflows. Our system demonstrates these features in the drug-discovery domain.

The main strength of the project is that it is structured and testable. The knowledge base contains 25 datasets, the benchmark includes 30 questions, and the system produces logs for memory and orchestration. This makes the project easier to evaluate. Another strength is that the system uses PubChem for real chemical information rather than guessing chemical properties.

A limitation of the project is that the default no-model mode may produce less natural responses than a full LLM-based chatbot. However, it is more reliable for evaluation because it can run without downloading a

large model file. Another limitation is that the project is an educational prototype and should not be used for real medical or laboratory decisions.

13. Conclusion

This project successfully developed a drug-discovery AI agent with structured knowledge retrieval, persistent memory, dynamic orchestration, and PubChem skill integration. The system includes 25 knowledge datasets and 30 benchmark questions. It passed validation with no errors and achieved an average benchmark score of 0.889, with all benchmark questions passing the evaluation threshold.

The project demonstrates how AI-agent architecture can be applied to a real life-sciences domain. It also shows how memory, tool use, and orchestration can make a system more advanced than a basic chatbot. Overall, the project satisfies the assignment requirements by providing complete code, documentation, evaluation benchmark results, and demo logs.

References

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Appendix A: GitHub Submission Summary

GitHub Project Folder:

https://github.com/BoolyLutoonom/ai-agent-assignment/tree/main/DL_Assignment

Selected Topic:

drug-discovery

Selected Optional Modules:

Multi-level Persistent Memory System
Advanced Dynamic Orchestration
PubChem MCP/SKILL Integration

Quick Test Results:

```
{
  "validation": {
    "knowledge_sets": 25,
    "benchmark_questions": 30,
    "errors": []
  },
  "benchmark_evaluation": {
    "question_count": 30,
    "average_score": 0.889,
    "passed_at_0_34": 30
  },
  "memory_demo": {
    "working_turns": 3,
    "semantic_entities": 4,
    "semantic_relations": 4,
    "status": "persistent memory recall working"
  },
  "orchestration_demo": {
    "status": "reports/orchestration_demo_traces.json generated"
  }
}
```

Model Note:

The Phi-3-mini-4k-instruct.Q4_0 GGUF model file is not included in the GitHub repository because it is too large. The project runs by default in retrieval-based mode. Optional local LLM mode can be enabled by placing the GGUF model file inside the `models/` folder.